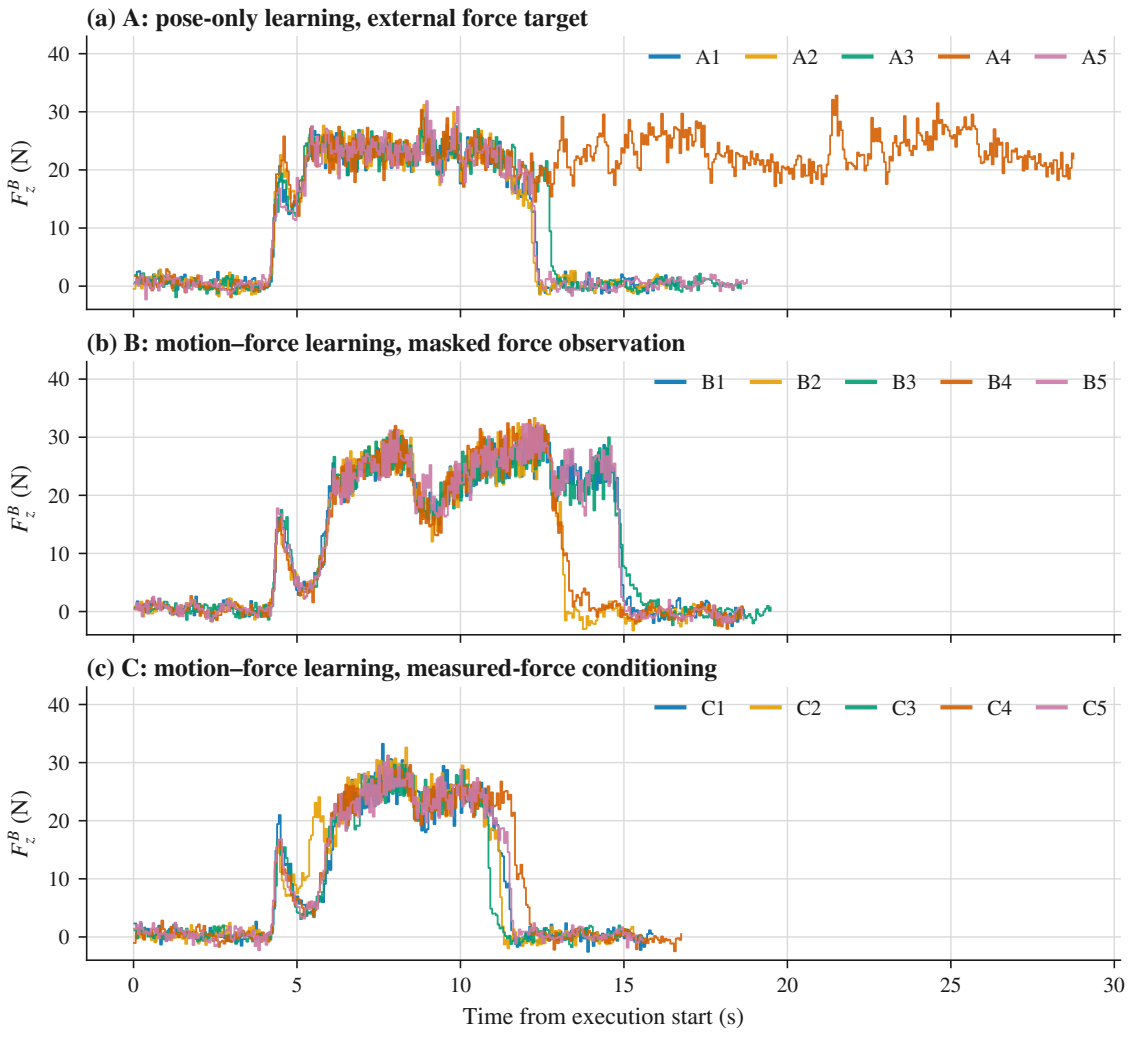

(a) A: pose-only learning, external force target
A1
A2
A3
A4
A5
(b) B: motion–force learning, masked force observation
B1
B2
B3
B4
B5
(c) C: motion–force learning, measured-force conditioning
C1
C2
C3
C4
C5
F_z^B (N)
0
10
20
30
40
0
5
10
15
20
25
30
Time from execution start (s)